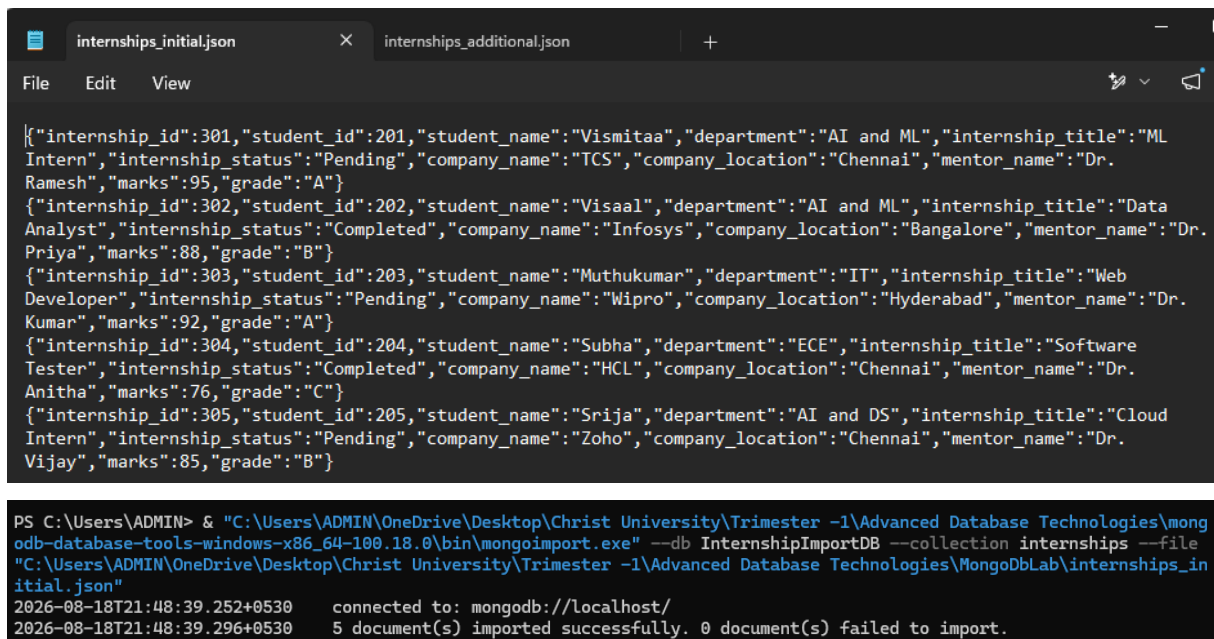


Exercise: Demonstration of Import, Export, Dump, and Restore Operations in MongoDB

With respect to your chosen business domain, create a MongoDB database and design the required collection(s). Prepare the dataset in either JSON format or CSV format based on the requirements of your selected domain.

Perform the following operations:

1. Import the prepared dataset into the appropriate MongoDB collection using the mongoimport utility.



```
internships_initial.json  internships_additional.json  +
File Edit View

[{"internship_id":301,"student_id":201,"student_name":"Vismitaa","department":"AI and ML","internship_title":"ML Intern","internship_status":"Pending","company_name":"TCS","company_location":"Chennai","mentor_name":"Dr. Ramesh","marks":95,"grade":"A"}
{"internship_id":302,"student_id":202,"student_name":"Visaal","department":"AI and ML","internship_title":"Data Analyst","internship_status":"Completed","company_name":"Infosys","company_location":"Bangalore","mentor_name":"Dr. Priya","marks":88,"grade":"B"}
{"internship_id":303,"student_id":203,"student_name":"Muthukumar","department":"IT","internship_title":"Web Developer","internship_status":"Pending","company_name":"Wipro","company_location":"Hyderabad","mentor_name":"Dr. Kumar","marks":92,"grade":"A"}
{"internship_id":304,"student_id":204,"student_name":"Subha","department":"ECE","internship_title":"Software Tester","internship_status":"Completed","company_name":"HCL","company_location":"Chennai","mentor_name":"Dr. Anitha","marks":76,"grade":"C"}
{"internship_id":305,"student_id":205,"student_name":"Srija","department":"AI and DS","internship_title":"Cloud Intern","internship_status":"Pending","company_name":"Zoho","company_location":"Chennai","mentor_name":"Dr. Vijay","marks":85,"grade":"B"}

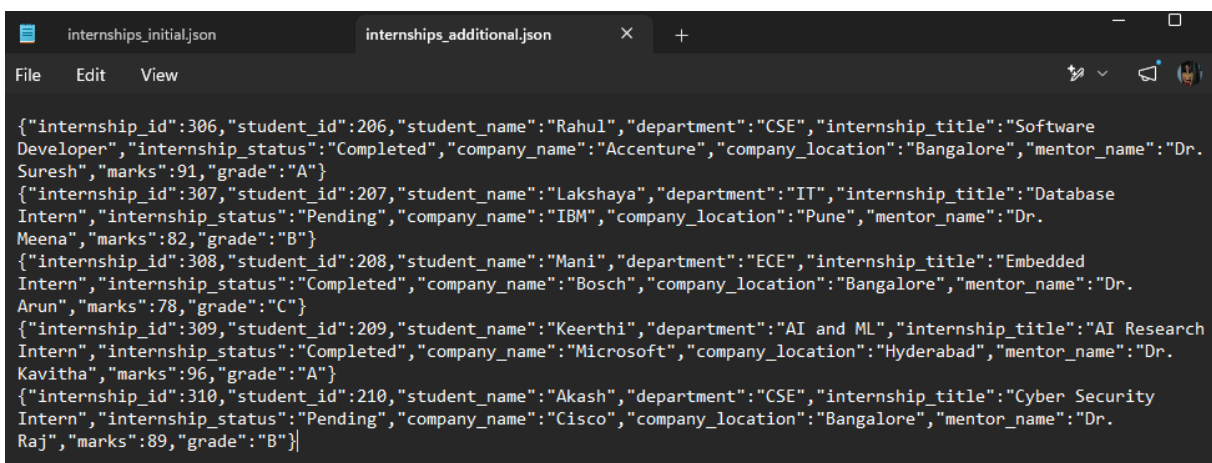
PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongoimport.exe" --db InternshipImportDB --collection internships --file "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\internships_initial.json"
2026-08-18T21:48:39.252+0530    connected to: mongodb://localhost/
2026-08-18T21:48:39.296+0530    5 document(s) imported successfully. 0 document(s) failed to import.
```

```

> use InternshipImportDB
< switched to db InternshipImportDB
> db.internships.countDocuments()
< 10
> db.internships.findOne()
< {
  _id: ObjectId('6a8485df3ec4e61e1fc3c0bf'),
  internship_id: 303,
  student_id: 203,
  student_name: 'Muthukumar',
  department: 'IT',
  internship_title: 'Web Developer',
  internship_status: 'Pending',
  company_name: 'Wipro',
  company_location: 'Hyderabad',
  mentor_name: 'Dr. Kumar',
  marks: 92,
  grade: 'A'
}

```

2. Append Additional Records: Create a separate JSON or CSV file containing at least five additional records relevant to the selected business domain. Import the file into the existing collection to append the new records.



```

internships_additional.json
File Edit View
{"internship_id":306,"student_id":206,"student_name":"Rahul","department":"CSE","internship_title":"Software Developer","internship_status":"Completed","company_name":"Accenture","company_location":"Bangalore","mentor_name":"Dr. Suresh","marks":91,"grade":"A"}
{"internship_id":307,"student_id":207,"student_name":"Lakshaya","department":"IT","internship_title":"Database Intern","internship_status":"Pending","company_name":"IBM","company_location":"Pune","mentor_name":"Dr. Meena","marks":82,"grade":"B"}
{"internship_id":308,"student_id":208,"student_name":"Mani","department":"ECE","internship_title":"Embedded Intern","internship_status":"Completed","company_name":"Bosch","company_location":"Bangalore","mentor_name":"Dr. Arun","marks":78,"grade":"C"}
{"internship_id":309,"student_id":209,"student_name":"Keerthi","department":"AI and ML","internship_title":"AI Research Intern","internship_status":"Completed","company_name":"Microsoft","company_location":"Hyderabad","mentor_name":"Dr. Kavitha","marks":96,"grade":"A"}
{"internship_id":310,"student_id":210,"student_name":"Akash","department":"CSE","internship_title":"Cyber Security Intern","internship_status":"Pending","company_name":"Cisco","company_location":"Bangalore","mentor_name":"Dr. Raj","marks":89,"grade":"B"}

```

```
> db.internships.find({
  internship_id: {$gte: 306}
})
< {
  _id: ObjectId('6a848635e6f30e85edb27f69'),
  internship_id: 306,
  student_id: 206,
  student_name: 'Rahul',
  department: 'CSE',
  internship_title: 'Software Developer',
  internship_status: 'Completed',
  company_name: 'Accenture',
  company_location: 'Bangalore',
  mentor_name: 'Dr. Suresh',
  marks: 91,
  grade: 'A'
}
```

3. Verify the data in the collection by performing the following:
 - a. Display the total number of documents in the collection.

```
PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongoimport.exe" --db InternshipImportDB --collection internships --file "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\internships_additional.json"
2026-08-18T21:50:05.450+0530    connected to: mongodb://localhost/
2026-08-18T21:50:05.455+0530    5 document(s) imported successfully. 0 document(s) failed to import.
PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongoexport.exe" --db InternshipImportDB --collection internships --out "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\internships_export.json"
2026-08-18T21:56:29.919+0530    connected to: mongodb://localhost/
2026-08-18T21:56:29.929+0530    exported 10 records
```

- b. Display a sample document from the collection.

```
{
  _id: ObjectId('6a848635e6f30e85edb27f6a'),
  internship_id: 309,
  student_id: 209,
  student_name: 'Keerthi',
  department: 'AI and ML',
  internship_title: 'AI Research Intern',
  internship_status: 'Completed',
  company_name: 'Microsoft',
  company_location: 'Hyderabad',
  mentor_name: 'Dr. Kavitha',
  marks: 96,
  grade: 'A'
}
```

c. Display the newly added records.

4. Export the documents from the collection into a JSON or CSV file using the mongoexport utility.

```
PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongoexport.exe" --db InternshipImportDB --collection internships --out "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\internships_export.json"
2026-08-18T21:56:29.919+0530    connected to: mongod://localhost/
2026-08-18T21:56:29.929+0530    exported 10 records
```

Name	Status	Date modified	Type	Size
 internships.bson		18/08/2026 21:57	BSON File	3 KB
 internships.metadata		18/08/2026 21:57	JSON Source File	1 KB
 prelude		18/08/2026 21:57	JSON Source File	1 KB

5. Create a complete backup of the MongoDB database using the mongodump utility.

```
PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongodump.exe" --db InternshipImportDB --out "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup"
2026-08-18T21:57:17.588+0530    writing 'InternshipImportDB.internships' to 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\internships.bson'
2026-08-18T21:57:17.597+0530    done dumping 'InternshipImportDB.internships' (10 documents)
```

6. Delete selected documents or drop the collection/database to demonstrate the need for data restoration.

```
> db.internships.deleteMany({
  internship_status: "Pending"
})
< {
  acknowledged: true,
  deletedCount: 5
}
> db.internships.countDocuments()
< 5
```

7. Restore the backed up database using the mongorestore utility.

```
PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongorestore.exe" --db InternshipImportDB "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB"
2026-08-18T22:24:04.804+0530 The --db and --collection flags are deprecated for this use-case; please use --nsInclude instead, i.e. with --nsInclude=${DATABASE}.${COLLECTION}
2026-08-18T22:24:04.807+0530 building a list of collections to restore from 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB' dir
2026-08-18T22:24:04.808+0530 don't know what to do with file 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\prelude.json', skipping...
2026-08-18T22:24:04.808+0530 reading metadata for 'InternshipImportDB.internships' from 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\internships.metadata.json'
2026-08-18T22:24:04.810+0530 restoring to existing collection 'InternshipImportDB.internships' without dropping
2026-08-18T22:24:04.810+0530 restoring 'InternshipImportDB.internships' from 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\internships.bson'
2026-08-18T22:24:04.822+0530 continuing through error: E11000 duplicate key error collection: InternshipImportDB.internships index: _id_ dup key: { _id: ObjectId('6a8485df3ec4e61efc3c0c1') }
2026-08-18T22:24:04.822+0530 continuing through error: E11000 duplicate key error collection: InternshipImportDB.internships index: _id_ dup key: { _id: ObjectId('6a8485df3ec4e61efc3c0c3') }
2026-08-18T22:24:04.822+0530 continuing through error: E11000 duplicate key error collection: InternshipImportDB.internships index: _id_ dup key: { _id: ObjectId('6a848635e6f30e85edb27f69') }
2026-08-18T22:24:04.822+0530 continuing through error: E11000 duplicate key error collection: InternshipImportDB.internships index: _id_ dup key: { _id: ObjectId('6a848635e6f30e85edb27f6a') }
2026-08-18T22:24:04.822+0530 continuing through error: E11000 duplicate key error collection: InternshipImportDB.internships index: _id_ dup key: { _id: ObjectId('6a848635e6f30e85edb27f6d') }
2026-08-18T22:24:04.823+0530 finished restoring 'InternshipImportDB.internships' (5 documents, 5 failures)
2026-08-18T22:24:04.823+0530 no indexes to restore for collection 'InternshipImportDB.internships'
2026-08-18T22:24:04.823+0530 5 document(s) restored successfully. 5 document(s) failed to restore.

PS C:\Users\ADMIN> & "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\mongodb-database-tools-windows-x86_64-100.18.0\bin\mongorestore.exe" --drop --db InternshipImportDB "C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB"
2026-08-18T22:24:37.443+0530 The --db and --collection flags are deprecated for this use-case; please use --nsInclude instead, i.e. with --nsInclude=${DATABASE}.${COLLECTION}
2026-08-18T22:24:37.445+0530 building a list of collections to restore from 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB' dir
2026-08-18T22:24:37.446+0530 don't know what to do with file 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\prelude.json', skipping...
2026-08-18T22:24:37.446+0530 reading metadata for 'InternshipImportDB.internships' from 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\internships.metadata.json'
2026-08-18T22:24:37.448+0530 dropping collection 'InternshipImportDB.internships' before restoring
2026-08-18T22:24:37.487+0530 restoring 'InternshipImportDB.internships' from 'C:\Users\ADMIN\OneDrive\Desktop\Christ University\Trimester -1\Advanced Database Technologies\MongoDbLab\InternshipBackup\InternshipImportDB\internships.bson'
2026-08-18T22:24:37.498+0530 finished restoring 'InternshipImportDB.internships' (10 documents, 0 failures)
2026-08-18T22:24:37.499+0530 no indexes to restore for collection 'InternshipImportDB.internships'
2026-08-18T22:24:37.499+0530 10 document(s) restored successfully. 0 document(s) failed to restore.
```

8. Verify that the database and collection have been restored successfully by displaying:
 - a. The list of databases.
 - b. The list of collections in the restored database.
 - c. The total number of documents in the restored collection.

```

test> exit

C:\Users\ADMIN>mongosh
Current Mongosh Log ID: 6a848f044c1d27409996dfff2
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.10.0
Using MongoDB:      8.3.7
Using Mongosh:       2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-08-17T15:05:21.353+05:30: Access control is not enabled for the database. Read and write access to data and conf
  igation is unrestricted
-----

test> show dbs
InternshipImportDB   40.00 KiB
InternshipManagement 236.00 KiB
admin                40.00 KiB
config               100.00 KiB
local                68.00 KiB
test> use InternshipImportDB
| show collections
switched to db InternshipImportDB
InternshipImportDB> db.internships.countDocuments()
10

```

- c. A sample document from the restored collection.

```

InternshipImportDB> db.internships.findOne()
{
  _id: ObjectId('6a8485df3ec4e61e1fc3c0bf'),
  internship_id: 303,
  student_id: 203,
  student_name: 'Muthukumar',
  department: 'IT',
  internship_title: 'Web Developer',
  internship_status: 'Pending',
  company_name: 'Wipro',
  company_location: 'Hyderabad',
  mentor_name: 'Dr. Kumar',
  marks: 92,
  grade: 'A'
}

```

9. Create an index on a relevant field in the collection based on the selected business domain. Execute a query that can benefit from the created index and verify the query execution using an appropriate MongoDB command.

```

,
InternshipImportDB> db.internships.createIndex({company_name: 1})
company_name_1
InternshipImportDB> db.internships.find({company_name: "TCS"})
[
  {
    _id: ObjectId('6a8485df3ec4e61e1fc3c0c0'),
    internship_id: 301,
    student_id: 201,
    student_name: 'Vismithaa',
    department: 'AI and ML',
    internship_title: 'ML Intern',
    internship_status: 'Pending',
    company_name: 'TCS',
    company_location: 'Chennai',
    mentor_name: 'Dr. Ramesh',
    marks: 95,
    grade: 'A'
  }
]

```

```

InternshipImportDB> db.internships.find({company_name: "TCS"}).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'InternshipImportDB.internships',
    parsedQuery: { company_name: { '$eq': 'TCS' } },
    indexFilterSet: false,
    queryHash: 'DE816D97',
    planCacheShapeHash: 'DE816D97',
    planCacheKey: '639CA21F',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'InternshipImportDB.internships',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'InternshipImportDB.internships',
        keyPattern: { company_name: 1 },
        indexName: 'company_name_1',
        isMultiKey: false,
        multiKeyPaths: { company_name: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,

```

```
    indexVersion: 2,
    direction: 'forward',
    indexBounds: { company_name: [ '["TCS", "TCS"]' ]
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 30,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'InternshipImportDB.internships',
    docsExamined: 1,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
```



```

    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'InternshipImportDB.internships',
    keyPattern: { company_name: 1 },
    indexName: 'company_name_1',
    isMultiKey: false,
    multiKeyPaths: { company_name: [] },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: { company_name: [ '['TCS', 'TCS']' ] },
    keysExamined: 1,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0,
    peakTrackedMemBytes: 0
  }
}
},

```

```

queryShapeHash: '21E6C83C480DF75BB42EB479CB73DBB2CD919592A39C467696AB567B3EB172C1',
command: {
  find: 'internships',
  filter: { company_name: 'TCS' },
  '$db': 'InternshipImportDB'
},
serverInfo: {
  host: 'DESKTOP-PK715JQ',
  port: 27017,
  version: '8.3.7',
  gitVersion: '34eee04f34989abb7a3d91447976f033f4f74af2'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```